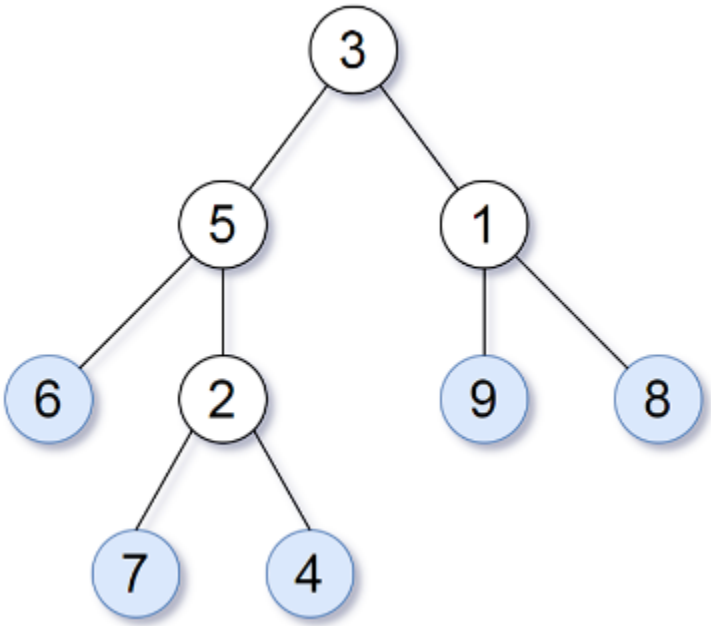
872. Leaf-Similar Trees

Problem Description:

Consider all the leaves of a binary tree, from left to right order, the values of those leaves form a **leaf value sequence.**

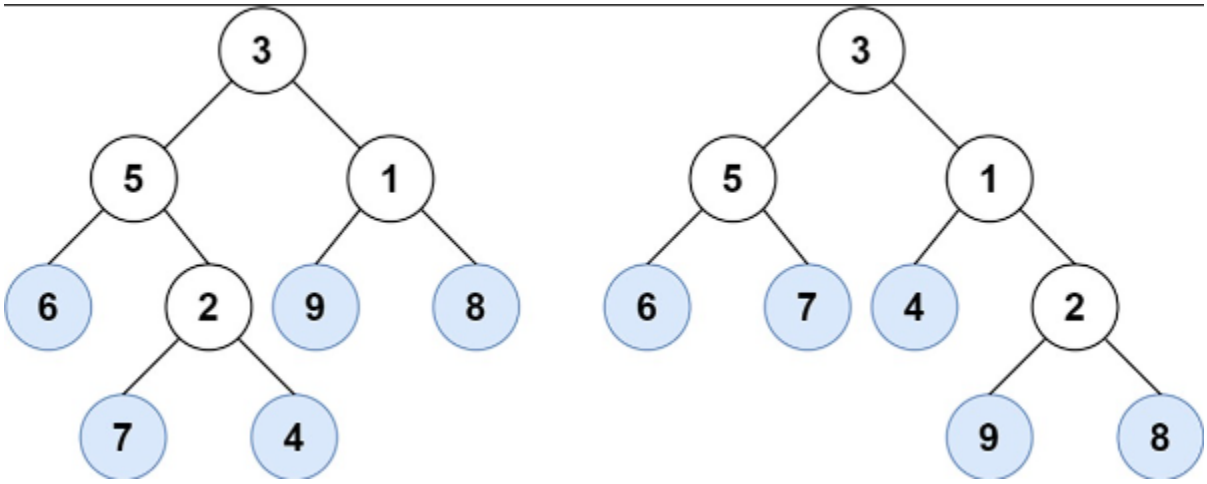


For example, in the given tree above, the leaf value sequence is (6, 7, 4, 9, 8).

Two binary trees are considered leaf-similar if their leaf value sequence is the same.

Return true if and only if the two given trees with head nodes root1 and root2 are leaf-similar.

Example 1:

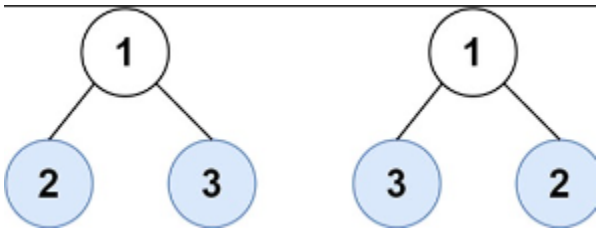


Input:

root1 = [3,5,1,6,2,9,8,null,null,7,4]
root2 = [3,5,1,6,7,4,2,null,null,null,null,null,null,9,8]

Output: true

Example 2:



Input:

root1 = [1,2,3],
root2 = [1,3,2]

Output: false

Constraints:

The number of nodes in each tree will be in the range [1, 200].
Both of the given trees will have values in the range [0, 200].

Solution:

```cpp
class Solution {
public:
    bool leafSimilar(TreeNode* root1, TreeNode* root2) {
        vector<int> leaves1, leaves2;

        getLeaf(root1, leaves1);
        getLeaf(root2, leaves2);

        if(leaves1 == leaves2)
            return true;
        return false;

    }

    void getLeaf(TreeNode* root, vector<int>& leaves) {
        if(!root)
            return;

        if(!root->left && !root->right) {
            leaves.push_back(root->val);
            return;
        }

        getLeaf(root->right, leaves);
        getLeaf(root->left, leaves);
    }
};
```

Discussion:

I created a recursive function called getLeaf that collects all the leaf nodes of a tree. A leaf is any node that has no left and right child. When a leaf is found, its value is added to a vector.

In the main function, I prepared two vectors to store the leaf values from the two trees. I then called the getLeaf function for each tree to fill these vectors.

After gathering all the leaf values, I compared the two vectors. If both vectors contain the same values in the same order, the trees are considered leaf-similar, so the result is true. Otherwise, the result is false.

Developer:
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